RAJ KUMAR PULGARI

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SUMMARY

- Data Analyst with **4+ years of experience** in data analytics, business intelligence, ETL development, and machine learning across airline and enterprise domains.
- Proficient in **Python, SQL (PostgreSQL, MongoDB), R, Power BI, Tableau, and Excel** for data analysis, visualization, and reporting.
- Experienced in building **interactive KPI dashboards** and automated reporting solutions that improve operational visibility and business decision-making.
- Strong expertise in designing and optimizing **ETL pipelines**, data cleaning, transformation, and integrating structured and unstructured datasets.
- Hands-on experience developing **machine learning, NLP, Generative AI (GenAI), and predictive analytics** solutions using Scikit-learn, TensorFlow, XGBoost, Flask, and Streamlit.
- Skilled in **AWS, Google Cloud Platform (GCP), Git/GitHub, Jupyter Notebook, and Agile methodologies** for developing scalable, cloud-based analytics solutions.
- Proven ability to perform **statistical analysis, forecasting, hypothesis testing, A/B testing, root cause analysis, and business intelligence reporting** to support strategic initiatives.
- Currently pursuing a **Master of Science in Computer Science** while leveraging strong analytical, problem-solving, and cross-functional collaboration skills to deliver data-driven business solutions.

SKILLS

Programming Languages | Python, R, SQL (PostgreSQL, MongoDB), Bash.
Data Visualization & BI Tools | Power BI, Tableau, MS Excel (Pivot Tables, Power Query), Matplotlib, Seaborn.
Data Management & ETL | Google Cloud, ETL Pipelines, Structured & Unstructured Data Integration, Data Modeling, Data Cleaning & Transformation.
Machine Learning & AI | Scikit-Learn, TensorFlow, XGBoost, NLP, Generative AI (GenAI), Large Language Models (LLMs), Prompt Engineering, Classification Models, Regression, Statistical Forecasting.
Analytics & Reporting | KPI Dashboards, Operational Reporting, Ad-hoc Analysis, Trend Analysis, What-if Simulations, Business Intelligence.
Testing & Validation | Hypothesis Testing, Root Cause Analysis, A/B Testing, Model Evaluation, Performance Benchmarking.
Cloud & Infrastructure | AWS, Google Cloud Platform (GCP), Data Pipelines, Cloud-hosted Analytics, Scalable Data Pipelines.
Project Tools & Version Control | Git, GitHub, Jupyter Notebook, Agile Methodologies, Flask, Streamlit.

EXPERIENCE

Data Analyst — **Mar 2025 – Present**
Southwest Airlines — **Dallas, TX**

- Developed an enterprise airline analytics solution for Southwest Airlines using Python, SQL, Power BI, and AWS to analyze customer booking trends, flight delays, cancellation patterns, and operational efficiency metrics.
- Built interactive KPI dashboards to monitor on-time performance, customer satisfaction trends, route profitability, and baggage handling metrics, improving reporting efficiency by 30%.
- Designed scalable ETL pipelines to process structured and semi-structured operational datasets, enabling automated reporting and near real-time business insights.
- Applied machine learning models and statistical forecasting techniques to predict flight delay risks, optimize resource allocation, and improve operational planning across airline routes.
- Utilized NLP and Generative AI techniques to analyze customer feedback and sentiment data, identifying recurring service issues and improving customer experience insights.
- Collaborated with cross-functional business teams in an Agile environment using Git/GitHub and Jupyter Notebooks to maintain scalable analytics pipelines and reproducible reporting workflows.

Data Analyst — **Jan 2021 – Jan 2024**
Cognizant — **Hyderabad, India**

- Analyzed and transformed large-scale structured and unstructured datasets using SQL (PostgreSQL, MongoDB) and Python (Pandas, NumPy) to generate actionable insights that supported strategic business decisions.
- Designed and maintained interactive dashboards using Power BI and Tableau to monitor KPIs, streamline operational reporting, and enhance data visibility for cross-functional stakeholders.
- Developed and deployed advanced machine learning models including classification, NLP, and GenAI using Scikit-learn, TensorFlow, Flask, and Streamlit to provide predictive analytics aligned with business goals.
- Led data preparation and transformation efforts through ETL pipeline development and advanced data cleaning techniques, reducing data processing time by 50%.
- Partnered with AI engineers and business stakeholders to build scalable data models and analytics frameworks supporting regression analysis, statistical forecasting, and hypothesis testing.
- Applied statistical techniques and visualization libraries (Matplotlib, Seaborn) to identify trends, detect anomalies, and deliver compelling data stories to executive audiences.
- Authored reproducible documentation for data workflows, machine learning pipelines, and performance metrics using Jupyter Notebooks, improving model transparency and scalability across teams.

ACADEMIC PROJECTS

Defense Acquisition Modernization Dashboard

- Developed a real-time analytics dashboard using Python, SQL, and Power BI to visualize defense procurement efficiency, contractor compliance, and policy impacts.
- Engineered an automated ETL pipeline for structured data ingestion and applied NLP techniques to extract insights from unstructured policy documents.
- Conducted comparative policy analysis through cost-risk modeling and what-if simulations, identifying inefficiencies and recommending changes projected to reduce delays by 23%.

AI Chatbot for Support

- Analyzed chat data using Python, SQL, and Power BI, improving chatbot intent recognition accuracy by 35% and reducing escalations by 25%.
- Designed automated reporting pipelines and weekly trend summaries to optimize chatbot performance and enhance user experience.
- Cleaned and structured unstructured chat logs with Excel (Power Query, Pivot Tables) and presented actionable insights via PowerPoint, decreasing data processing time by 40%.

Traffic Accident Severity Prediction

- Built and validated machine learning models (XGBoost, Scikit-learn, R) achieving 85% prediction accuracy for accident severity classification.
- Integrated Flask and Google Maps API for real-time predictions and managed data pipelines using SQL, MongoDB, AWS, and Google Cloud.
- Automated data preprocessing and maintained codebase using Git/GitHub, ensuring scalable and reproducible analytics workflows.

EDUCATION

Master of Science in Computer Science, Stevens Institute of Technology, Hoboken, NJ
Dec 2025

Coursework: Data Analytics, Machine Learning, Database Management System, Python Programming, UI Designing & Interface, Data Transmission & API Access, Agile Frameworks, Cyber Security.

Bachelor of Technology in Information Technology Mahatma Gandhi Institute of Technology**,** Hyderabad, INDIA
Oct 2022

Coursework: Data Structures and Algorithms, C++, JAVA Programming, Data Communication & Computer Networks, Clustering & Neural Networks.